

How Far Can Lightweight Text Encoders Go in End-of-Turn Detection?

A Comparison of BoW, BiLSTM, and DistilBERT on FTAD

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Abstract

Spoken-dialogue systems must decide when a user has finished speaking before they respond. This report compares three text-only encoders on end-of-turn (EOT) detection: bag-of-words with a feedforward network (BoW+FFN), a bidirectional LSTM (BiLSTM), and fine-tuned DistilBERT. All models use up to three prior dialog turns plus the current utterance. Training uses the full FTAD split (119,696 examples); evaluation covers the held-out test set (14,961) and zero-shot transfer to TURNS-2K (2,000).

On FTAD with normalized text and context enabled, macro-F1 rises from 0.638 (BoW) to 0.677 (BiLSTM) to 0.700 (DistilBERT). BiLSTM reaches about 97% of DistilBERT’s in-domain score at roughly 10% of the parameters and seven times faster training. DistilBERT adds only +2.3 percentage points in-domain, but stays stable under raw vs. normalized text while BoW drops 8.0 pp on raw input. Three-turn context does not help any encoder. On TURNS-2K, lightweight models degrade sharply; DistilBERT reaches 0.749 macro-F1. Pre-training mainly helps generalization, not the small in-domain gap over BiLSTM.

1 Introduction

End-of-turn detection sits between speech recognition and downstream dialog logic: respond too early and the system interrupts; respond too late and the interaction feels slow. Recent systems combine audio, voice-activity prediction, and large language models, but text-only classifiers still matter when transcripts already exist and latency must stay low.

The practical question is not whether DistilBERT wins on accuracy alone, but whether lighter models are good enough for fixed-domain pipelines. This report uses *lightweight* for BoW+FFN and BiLSTM trained from scratch, and *pretrained* for full fine-tuning of DistilBERT-base. Audio, frame-wise voice-activity prediction, and generative turn detectors are out of scope. The goal is a controlled encoder comparison aligned with HHU course methods (Practicals 4, 5, and 8), not a state-of-the-art replication of Chen et al.’s RoBERTa+BiGRU system.

Research question: How far can lightweight text encoders go in end-of-turn detection, and where does a pretrained encoder still add measurable value?

We test five hypotheses on FTAD [1] with supporting zero-shot evaluation on TURNS-2K [5] (Table 1). The main comparison uses normalized text and three-turn context on the full training split. Additional runs cover context ablation, raw vs. normalized text, and out-of-domain transfer.

2 Background and Related Work

Chen et al. [1] introduce FTAD, a Switchboard-based dataset with fine-grained turn-taking labels. The `eot.txt` task predicts binary floor release after an inter-pausal unit (IPU). That label is related to, but not the same as, mere speaker change.

Table 1: Hypotheses and outcomes.

	Hypothesis	Outcome
H1	DistilBERT $\geq$ BiLSTM $>$ BoW (macro-F1)	Supported
H2	Diminishing returns along encoder ladder	Supported
H3	Three-turn context improves all encoders	Not supported
H4	Lightweight models more sensitive to normalization	Partial
H5	OOD performance $<$ in-domain	Partial

TurnGPT [2] models turn shifts generatively with GPT-2, but inference cost is higher than a small classifier. Chen et al. report strong FTAD results with RoBERTa+BiGRU ($\sim 68\%$ on comparable splits); we use that number as reference only.

Anthus [3] shows that turn detectors trained on punctuated ASR output can exploit trailing punctuation when it reflects post-editing rather than online prosody. We therefore compare *raw* transcripts with *normalized* text (lowercase, punctuation removed). We do not assume raw text must outperform normalized text: for BoW, raw input can increase token sparsity; for pretrained encoders we expect more stable performance across both conditions.

Zuluaga-Gomez et al. [4] review heterogeneous EOT definitions and domain dependence, which motivates zero-shot evaluation on TURNS-2K. TurnSense [7] fine-tunes SmolLM2 on TURNS-2K and reports $\sim 97.5\%$ in-domain accuracy; we do not replicate that setup, but it provides an upper-bound reference for the same benchmark under in-domain training.

3 Data and Preprocessing

3.1 In-domain: FTAD-sw

We use the public FTAD-sw release (`eot.txt`) with official splits: train 119,696, validation 14,961, test 14,961 examples. Each line is tab-separated: up to three prior turns (`context`), the `subject_utterance`, and a binary EOT label. We cite the FTAD paper and repository [1, 6] and do not redistribute raw files (no explicit open-source license).

Validation is $\sim 59\%$ positive (EOT=1). A majority-class baseline that always predicts EOT reaches 59.3% accuracy but only 0.372 macro-F1.

3.2 Out-of-domain: TURNS-2K

TURNS-2K [5] (Apache 2.0) contains 2,000 single utterances labeled complete vs. incomplete. We use it for zero-shot evaluation only: models trained on FTAD are applied without fine-tuning. TURNS-2K has no multi-turn context field, so inputs reduce to the utterance string.

3.3 Preprocessing

Two text modes are implemented in `preprocess.normalize()`: *raw* (strip whitespace only) and *normalized* (lowercase, remove punctuation, collapse whitespace). Normalized text is the primary condition for main results. With context enabled, prior turns are concatenated with the subject utterance (max length 128 tokens/subwords). BoW and BiLSTM vocabulary is built from training data only.

4 Methodology

All models output binary logits with cross-entropy loss. Training uses Adam with learning rate 10^{-3} for BoW and BiLSTM and 2×10^{-5} for DistilBERT, following course defaults for from-scratch vs. fine-tuned models. Batch sizes are 32/32/16; early stopping uses patience 2 on validation

Table 2: Model architectures (full-data runs).

Model	Encoder	Head	Params
BoW+FFN	Binary BoW (train vocab)	2-layer FFN, ReLU	8.8M
BiLSTM	Emb. 128; 2-layer BiLSTM, $h=256$	Linear on final state	6.8M
DistilBERT	<code>distilbert-base-uncased</code>	Linear on [CLS]	66.4M

macro-F1; seed 42; hardware is Apple M4 Pro with PyTorch MPS. Software: PyTorch, Hugging Face `transformers`, scikit-learn. Code, seeds, hyperparameters, and reproduction instructions are in github.com/Onursonn/Comparing-Text-Encoders-on-End-Of-Turn-Tasks-Paper.

Development ablations use stratified 20k subsets; **main results use the full training split**. Primary metric is macro-F1; we also report accuracy, precision/recall for EOT=1, confusion matrices, and inference latency on 1,000 FTAD test examples (batch mode, after warmup).

Limitations at setup time: context and raw/normalized ablations run on 20k subsets only; TURNS-2K labels differ from FTAD floor release; we use a single seed and no significance testing across reruns.

5 Results

5.1 Main comparison on full FTAD

Table 3 and Figure 1 show the primary experiment (normalized text, context on, full training).

Table 3: Main comparison on FTAD test (full training, normalized, context on).

Model	F1	Acc	Prec	Rec	Train	ms/ex
Majority	0.372	0.593	—	—	—	—
BoW+FFN	0.638	0.653	0.691	0.781	8.2m	1.15
BiLSTM	0.677	0.684	0.792	0.619	12.0m	0.43
DistilBERT	0.700	0.706	0.767	0.728	83.2m	2.85

DistilBERT leads (+2.3 pp over BiLSTM, +6.2 pp over BoW), supporting H1. The step BoW→BiLSTM (+3.9 pp) is larger than BiLSTM→DistilBERT (+2.3 pp), supporting H2. BoW is recall-heavy; BiLSTM is precision-heavy; DistilBERT is the most balanced (Figure 2). BiLSTM also has the lowest latency (0.43 ms/example) at ~97% of DistilBERT’s F1.

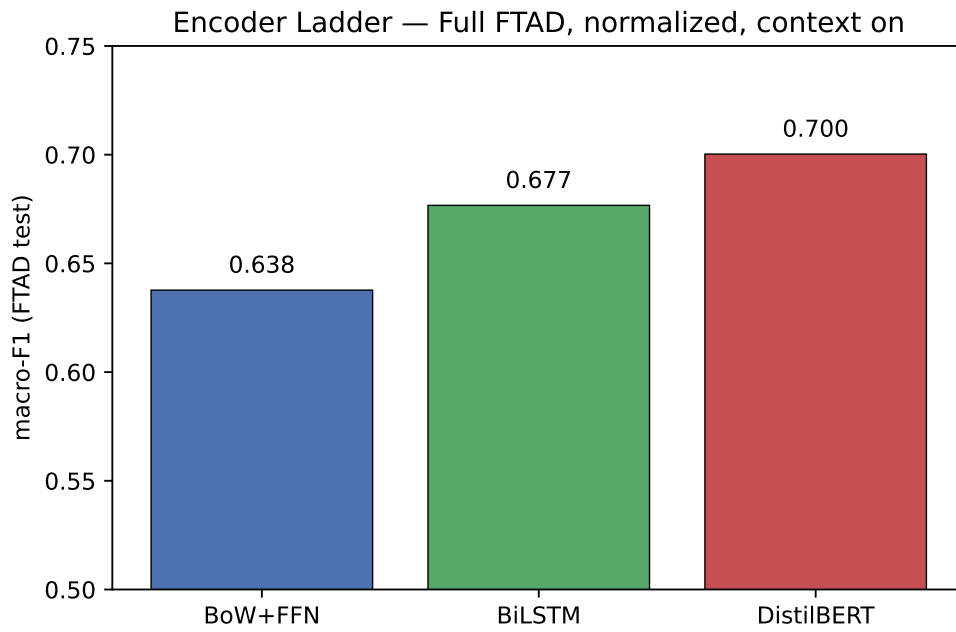


Figure 1: Macro-F1 on FTAD test for the three encoders.

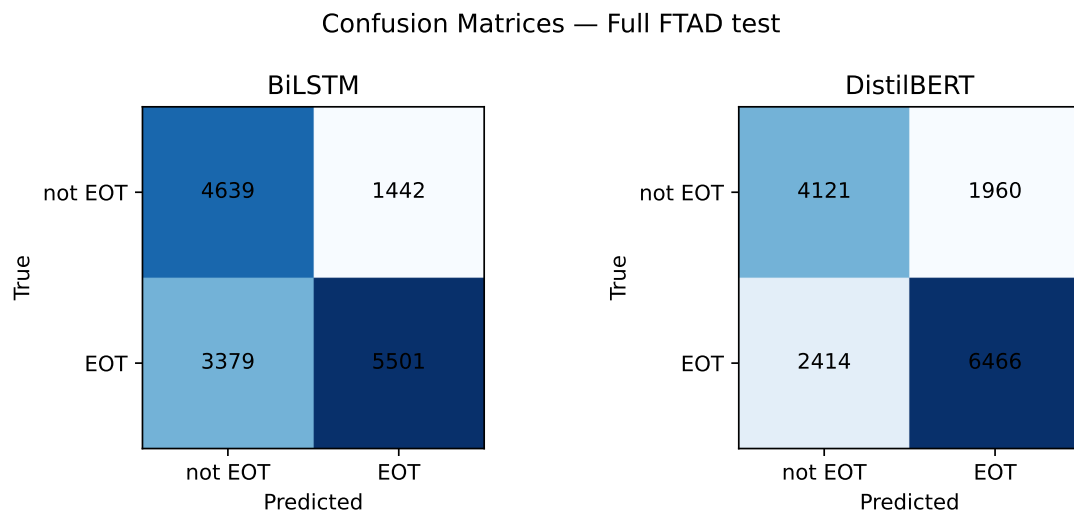


Figure 2: Confusion matrices on FTAD test: BiLSTM vs. DistilBERT.

5.2 Context ablation (20k subset)

Table 4 compares three-turn context against subject-only input. Context does not help; lightweight models perform slightly better without it (H3 not supported). DistilBERT is flat across both settings. Likely causes are redundant context relative to the current utterance, noise in prior turns, and truncation at 128 tokens.

Table 4: Context ablation on 20k subset (normalized).

Model	Context on	Context off	$\Delta F1$
BoW+FFN	0.635	0.664	+0.029
BiLSTM	0.614	0.660	+0.046
DistilBERT	0.688	0.687	−0.001

5.3 Raw vs. normalized text (distinction experiment)

Figure 3 compares training and evaluation on raw vs. normalized text (20k subset, context on).

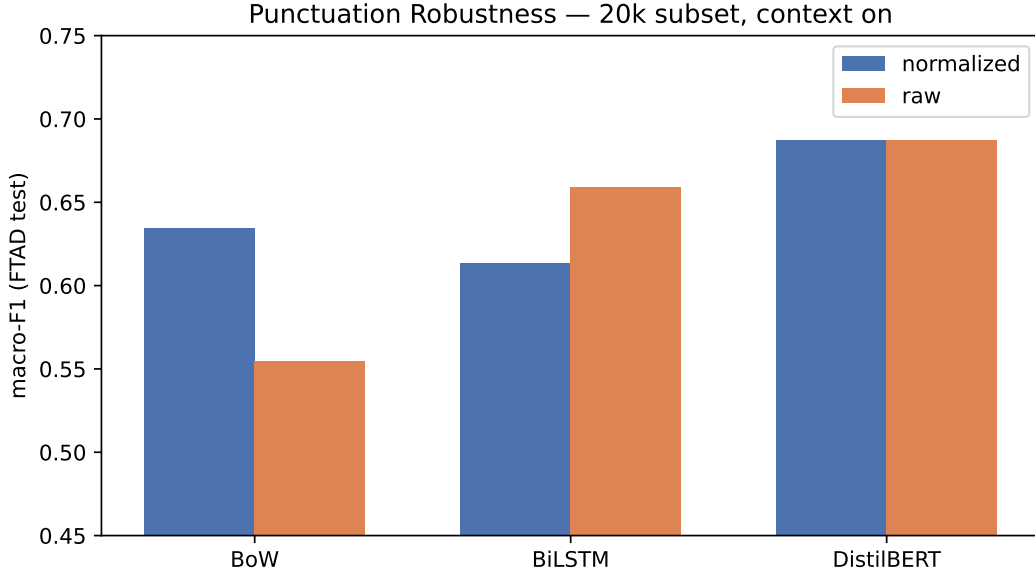


Figure 3: Macro-F1 on raw vs. normalized text (20k subset). BoW drops on raw input; DistilBERT is stable.

BoW falls from 0.635 to 0.555 (−8.0 pp) on raw text. That pattern does not fit punctuation shortcuts: if BoW relied on trailing periods, raw should outperform normalized text. A more plausible explanation is vocabulary sparsity when case and punctuation are kept. DistilBERT stays at ~ 0.687 (H4 partially supported). BiLSTM raw > normalized on the subset is treated as a subset artifact.

5.4 Zero-shot evaluation on TURNS-2K

Table 5 applies full-data checkpoints to TURNS-2K without fine-tuning.

Lightweight models collapse out-of-domain (H5 for BoW and BiLSTM). DistilBERT improves to 0.749 macro-F1 despite different label definitions and shorter utterances. This is far below TurnSense’s in-domain $\sim 97.5\%$ on TURNS-2K [7], which underscores that our FTAD-trained models do not transfer cleanly even when the stronger encoder generalizes best.

Table 5: Zero-shot evaluation on TURNS-2K.

Model	FTAD F1	TURNS-2K F1	Δ
BoW+FFN	0.638	0.480	-0.158
BiLSTM	0.677	0.512	-0.165
DistilBERT	0.700	0.749	+0.049

5.5 Error analysis

We reviewed 10 false positives and 10 false negatives per model on the FTAD test set. Table 6 shows representative cases; patterns recur across models.

Table 6: Representative misclassifications on FTAD test (truncated).

Utterance	Gold	BoW	BiLSTM	DistilBERT	Pattern
um-hum	not EOT	EOT	EOT	EOT	backchannel
yeah	not EOT	EOT	not EOT	not EOT	backchannel
... I think but	not EOT	EOT	not EOT	not EOT	trailing disfluency
the reason is that we	not EOT	not EOT	not EOT	not EOT	incomplete clause
... and you know and	EOT	not EOT	EOT	EOT	disfluent completion

All three models confuse backchannels such as **um-hum** and **yeah**. BoW over-predicts EOT on trailing disfluencies (**but**, **in**). BiLSTM misses long incomplete clauses after normalization. DistilBERT makes fewer gross errors but still fails on disfluent completions.

6 Discussion and Conclusion

The data supports a split answer to the research question. For fixed-domain Switchboard-style transcripts with reliable normalization, BiLSTM is already usable (0.677 macro-F1, lowest latency) and DistilBERT adds only a small in-domain margin (+2.3 pp). DistilBERT pays off when preprocessing varies or the domain shifts: BoW loses 8 pp on raw text while DistilBERT stays flat, and BiLSTM loses 16.5 pp on TURNS-2K while DistilBERT gains 4.9 pp. Three-turn context, in this setup, does not improve any encoder.

- **Fixed-domain deployment:** BiLSTM is the practical default if transcripts are normalized and the domain matches training data.
- **Noisy ASR or domain shift:** DistilBERT is worth the extra compute and parameter cost.
- **Pipeline design:** invest in normalization before context windows; the null context result suggests limited return from three-turn history on FTAD with our input format.

Remaining limitations: ablations on 20k subsets only; TURNS-2K labels differ from FTAD floor release; single seed (42); no replication of Chen’s architecture; full DistilBERT training took ~ 83 min on M4 Pro MPS (< 2 h total project compute).

Lightweight encoders are not replaced by DistilBERT on FTAD, but the benefit of pretraining shows up mainly in robustness and transfer, not in the headline in-domain F1 gap.

References

- [1] Y. Chen, X. Wang, Y. Zhang, and J. Hirschberg. Human-to-human conversation dataset for learning fine-grained turn-taking action. *Interspeech*, 2021.

- [2] K. Ekstedt and G. Skantze. TurnGPT: A transformer-based language model for predicting turn-taking in spoken dialog. *Findings of EMNLP*, 2020.
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- [4] J. Zuluaga-Gomez et al. End-of-turn detection: A survey. *IWSds*, 2025.
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AI Usage Declaration

Experiment design, hypotheses, implementation, training, and evaluation were my own work. ChatGPT assisted throughout the project: brainstorming ideas, research, drafting report sections, and resolving coding difficulties. I expanded, rewrote, and verified all sections, code, and results myself. I take full responsibility for the final submission.